

Accessible Graphics

College of Charleston

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Today we will be discussing making graphics accessible

- 8% of males and .5% of females are colorblind
 - ▶ Red and Green colorblindness is more common than AB- blood type
- Some will be low hanging fruit
 - ▶ Color blind friendly color palettes
 - ▶ Using shapes/textures
 - ▶ Text and font decisions
- One will be more advanced
 - ▶ Alt text for graphics

Colorblind Types



NORMAL VISION



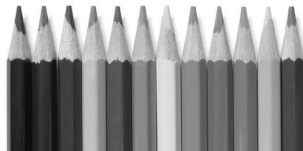
DEUTERANOMALIA



PROTANOPIA



TRITANOPIA



TOTAL COLOR BLINDNESS

Colorblind Challenges

Four main problems arise (as per [Okabe and Ito](#))

- ❶ Cannot distinguish certain colors
 - ▶ Red and green dots are going to fail and fail hard
- ❷ Certain colors become difficult to see
 - ▶ Dark red vanishes in dark backgrounds
- ❸ Similar to 2, things cannot be stressed in certain colors
 - ▶ Dark red text looks black and loses the emphasis
- ❹ Difficulty in naming colors
 - ▶ Detecting differences is different from naming colors!
 - ▶ Don't say "the red points"
 - ▶ Colorblind and normal vision create different naming boundaries

Colorblind Friendly Palettes

- Some color palettes are more accessible than others
 - ▶ High Contrast colors
 - ▶ [Ichihara et al](#) introduced a set of four color combinations that are optimized for colorblind people
- Base R went color palette friendlier circa 2019 ([announcement here](#))
- Classic advice: Avoid red and green at all costs
- Avoid yellow-green color spectrum generally
- Aesthetics are a thing that should still be respected
 - ▶ “Break any of these rules sooner than [make a data visualization] outright barbarous” -George Orwell
 - ▶ Eg Rainbows with Pride is fine

Non-Color Based Solutions

Redundant encoding for graphs is encoding the information in multiple ways (red circles, blue squares, green triangles, etc..)

- Examples include...
 - ▶ shapes for scatterplots
 - ▶ line types when used (solid-, dotted-, dashed-, etc..)
 - ▶ textures on bar charts
 - ▶ the brightness of the colors chosen (eg dark red vs light green)
- Avoid conveying information using only one medium

If you can convert the graph to greyscale and it still carries the same information, you are doing fine

Non-Color Based Solutions

- Label lines/symbols directly on graphs if plausible
 - ▶ Cuts down on trying to figure out which subway line is the “red” one
- Don't be too wordy with axis or labels
 - ▶ Leads to small font or cluttered graphics
 - ▶ Counters the goal of graphics being quick and easy to interpret
 - ▶ Needed with the above point
- Make lines/symbols extra thick
 - ▶ Easier to differentiate similar hues

- Don't use small fonts!!!
 - ▶ Absolute most common mess up with posters
 - ▶ 11 point font on a 36x48 poster is a terrible idea, terrible to read, and terribly easy to fix
 - ▶ Be less wordy
- Advice: Look at the graph yourself in the medium it'll be shown
 - ▶ Eg prof.s are encouraged to do dry runs of our presentations initially in the classrooms we teach to see how they look
- Avoid fonts with serfs (excess fancy things)
- This text uses a typeface that is suppose to be good for people with Dyslexia

Alternative (Alt) Text for Statistical Graphics

- Graphs offer information which can be lost for the blind or those with poor vision
- **Alt text** is effectively a verbose caption of a graph
- Usually not shown
- Best Practices:
 - ▶ Focus on what you are trying to convey with the graph
 - ▶ Mention the type of graph you are using (bar chart, scatterplot, etc...)
 - ▶ Imagine that you have the context of the image but can't see it; what would you like to know?
 - ▶ Don't duplicate text

Take Aways

Be conscious that color-blind, blind, and people with poor vision exist!

- Color-blind friendly color palettes
 - ▶ Avoid red + green and the green-yellow spectrum
- Redundant encoding (don't convey information in color only!)
- Readable fonts
- Avoid fonts with serifs
 - ▶ Dyslexic fonts exist if you know your audience
- Alt-text is a verbose description of graphs
 - ▶ Excellent if you want your graph seen by the world
- Aesthetics is a thing